

Astro HW 9

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9.1 Extinction

a) Apparent Mag In front

Given ...

$$m - M = 5 \log(d) - 5$$

$$M = -1.1$$

$$d = 800 \text{pc}$$

We find ...

$$m = 5 \log(d) - 5 - 1.1 \approx 8.415$$

b) Apparent Mag Behind Nebula

Given ...

$$m - M = 5 \log(d) - 5 + A_v$$

$$A_v = 1.1$$

$$d = 820$$

We find that ...

$$m = 5 \log(820) - 5 - 1.1 + 1.1 \approx 9.569$$

c) Error

It would be further than it actually is. This is because the stars' brightness would be dimmer, which would incorrectly imply that the star is further away.

Given ...

$$m = M + 5 \log(d) - 5$$

$$10^{(m-M)/5} = d$$

We find that ...

$$d = 10^{(9.569+1.1+5)/5} \approx 1360.8 \text{pc}$$

$$1360/820 \approx 1.65 \approx 65\%(\text{error})$$

9.2 Ionizing stars and HII regions

a)

Given ...

$$E = hv = hc/\lambda$$

9.3 Conservation of angular momentum

a) Deriving 17.15

Given ...

$$\omega = 2/5mr v$$

$$\omega_0 = 2/5mr_0 v_0$$

$$\omega_f = 2/5mr_f v_f$$

We find that ...

$$2/5mr_f v_f = 2/5mr_0 v_0$$

$$\cancel{2/5mr_f v_f} = \cancel{2/5mr_0 v_0}$$

$$v_f = \frac{r_0}{r_f} v_0$$

b) Verify Scaling

Given ...

$$\frac{r_0}{r_f} v_0 = v_f$$

$$\frac{GM}{r_f^2} = \frac{v_f^2}{r_f}$$

We find that...

$$\frac{GM}{r_f} = v_0^2 \frac{r_0^2}{r_f^2}$$

$$\cancel{\frac{1}{r_f}} r_f^2 = v_0^2 \frac{r_0^2}{GM}$$

$$r_f = \frac{v_0^2 r_0^2}{GM}$$

Given...

$$M = 2 \times 10^{30}$$

$$v = 100 \text{ m/s}$$

$$r = 10^{10} \text{ m}$$

$$G = 6.67 \times 10^{-11}$$

We find that...

$$r_f = \frac{100^2 \times 10^{10}}{6.67 \times 10^{-11} \times 2 \times 10^{30}}$$

$$r_f \approx \boxed{7.496 \times 10^{-15} \text{ m}}$$

c) Enhancement

Given ...

$$\omega_0 = 1/5mr_0v_0$$

$$\omega_f = 2/5mr_fv_f$$

We find that ...

$$2/5mr_fv_f = 1/5mr_0v_0$$

$$2/5mr_fv_f = 1/5mr_0v_0$$

$$\boxed{2\frac{r_0}{r_f}v_0 = v_f}$$

9.4 Post-MS nucleosynthesis

a) Triple-alpha Energy Release

Given ...

$$E = mc^2$$

$$M_{He} = 4.0026032497\text{u} \quad M_c = 12\text{u}$$

$$u = 1.66053873 \times 10^{-27}$$

We find that ...

$$\Delta M = 12\text{u} - 3 \times (M_{He} \times u) \approx -1.29685044 \times 10^{-29}$$

$$E_{rel} = (1.29685044 \times 10^{-29})(300000000)^2$$

$$E_{rel} \approx 1.167165 \times 10^{-12}$$

$$E_{pNuc} = \frac{1.167165 \times 10^{-12}}{12} \boxed{\approx 9.7263783 \times 10^{-14}\text{J}}$$

b) Star with He Core lifetime

Given ...

$$M_s = 0.3M_\odot = 0.3 \times 1.989 \times 10^{30} \approx 5.97 \times 10^{29} \text{ kg}$$

$$L = 25L_\odot = 25 \times 3.828 \times 10^{26} \approx 9.57 \times 10^{27} \text{ W}$$

$$E_{rxn} = 9.7263783 \times 10^{-14}\text{J}$$

$$m_{He} = M_{He} \times u = 6.64648 \times 10^{-27} \text{ kg}$$

We find that ...

$$N_{He} = \frac{M_s}{m_{He}} = 8.98 \times 10^{55}$$

$$N_{\text{rxn}} = \frac{N_{\text{He}}}{3} = 2.99 \times 10^{55}$$

$$E_{\text{tot}} = N_{\text{rxn}} E_{\text{rxn}} = 3.49 \times 10^{43} \text{ J}$$

$$t_{\text{HeBurn}} = \frac{E_{\text{tot}}}{L} = \frac{3.49 \times 10^{43}}{9.57 \times 10^{27}} \boxed{\approx 3.64 \times 10^{15} \text{ s} \approx 1.16 \times 10^8 \text{ yr}}$$

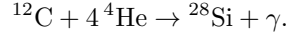
c) Energy released from C → Si and Si → Fe Fusion

Given ...

$$E = mc^2, \quad c = 3.00 \times 10^8 \text{ m s}^{-1}, \quad u = 1.66053873 \times 10^{-27} \text{ kg}$$

$$M_{\text{He}} = 4.0026032497 \text{ u}, \quad M_{\text{C}} = 12.000000 \text{ u}, \quad M_{\text{Si}} = 27.976926 \text{ u}, \quad M_{\text{Fe}} = 55.934942 \text{ u}.$$

(i) C → Si

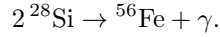


$$\Delta m_{\text{C} \rightarrow \text{Si}} = (M_{\text{C}} + 4M_{\text{He}}) - M_{\text{Si}} = 0.033486999 \text{ u}.$$

$$E_{\text{C} \rightarrow \text{Si}} = \Delta m_{\text{C} \rightarrow \text{Si}} c^2 = (0.033486999 \text{ u}) \left(1.66053873 \times 10^{-27} \frac{\text{kg}}{\text{u}} \right) (3.00 \times 10^8)^2 \approx 5.00 \times 10^{-12} \text{ J}.$$

$$E_n = \frac{E_{\text{C} \rightarrow \text{Si}}}{28} \approx 1.79 \times 10^{-13} \text{ J}.$$

(ii) Si → Fe



$$\Delta m_{\text{Si} \rightarrow \text{Fe}} = (2M_{\text{Si}}) - M_{\text{Fe}} = 0.018910 \text{ u}.$$

$$E_{\text{Si} \rightarrow \text{Fe}} = \Delta m_{\text{Si} \rightarrow \text{Fe}} c^2 = (0.018910 \text{ u}) \left(1.66053873 \times 10^{-27} \frac{\text{kg}}{\text{u}} \right) (3.00 \times 10^8)^2 = 2.83 \times 10^{-12} \text{ J}.$$

$$E_n = \frac{E_{\text{Si} \rightarrow \text{Fe}}}{56} = 5.05 \times 10^{-14} \text{ J}$$

As expected, the energy released *per nucleon* decreases as you go from C to Si to Fe.

9.5 Extra Credit

a) Deriving v_{dm}

Given...

$$Ke = 1/2mv^2, \quad GPE = -\frac{GMm}{r}$$

Because the object starts at rest...

$$Ke - GPE_f = -GPE_0$$

We find that ...

$$1/2mv_{dm}^2 - \frac{GMm}{r} = -\frac{GMm}{r_0}$$

$$1/2mv_{dm}^2 = -\frac{GMm}{r_0} + \frac{GMm}{r_0}$$

$$v_{dm}^2 = 2 \left(\frac{GMm}{mr} - \frac{GMm}{mr_0} \right)$$

$$v_{dm}^2 = 2 \left(\frac{GM\cancel{m}}{\cancel{m}r} - \frac{GM\cancel{m}}{\cancel{m}r_0} \right)$$

$$v_{dm} = \sqrt{2GM \left(\frac{1}{r} - \frac{1}{r_0} \right)}$$

Assuming in-fall means negative v...

$$v_{dm} = -\sqrt{2GM \left(\frac{1}{r} - \frac{1}{r_0} \right)}$$

b) Integrating $v(r)$ as dr/dt

Given ...

$$M = (4\pi/3)R_0^3\rho_0$$

We find ...

$$dr/dt = v_{dm} = -\sqrt{2GM\left(\frac{1}{r} - \frac{1}{r_0}\right)}$$

$$dt = \frac{dr}{-\sqrt{2GM\left(\frac{1}{r} - \frac{1}{r_0}\right)}}$$

$$t = \int_{R_0}^r \frac{1}{-\sqrt{2GM\left(\frac{1}{r} - \frac{1}{r_0}\right)}} dr$$

$$t = -\frac{1}{\sqrt{2GM}} \int_{R_0}^r \frac{1}{\sqrt{\left(\frac{1}{r} - \frac{1}{r_0}\right)}} dr$$

$$u = \frac{r}{r_0}, \quad du = 1/r_0 dr, \quad r_0 du = dr, \quad R_0 \rightarrow 0, \quad r \rightarrow 1$$

$$t = -\frac{r_0}{\sqrt{2GM}} \int_0^1 \frac{1}{\sqrt{\left(\frac{1}{ur_0} - \frac{1}{r_0}\right)}} du$$

$$t = -\frac{r_0\sqrt{r_0}}{\sqrt{2GM}} \int_0^1 \frac{1}{\sqrt{\left(\frac{1}{u} - \frac{1}{1}\right)}} du$$

$$t = -\frac{\sqrt{r_0^3}}{\sqrt{2GM}} \int_0^1 \frac{1}{\sqrt{\left(\frac{1-u}{u}\right)}} du$$

$$t = -\frac{\sqrt{r_0^3}}{\sqrt{2GM}} \int_0^1 \sqrt{\left(\frac{u}{1-u}\right)} du, \quad \int_0^1 \sqrt{\left(\frac{x}{1-x}\right)} dx = \frac{\pi}{2}$$

$$t = -\left(\frac{\sqrt{r_0^3}}{\sqrt{2GM}}\right) \left(\frac{\pi}{2}\right)$$